Machine Learning Overview

Foreword

- Machine learning is a core research field of AI, and it is also a necessary knowledge for deep learning. Therefore, this chapter mainly introduces the main concepts of machine learning, the classification of machine learning, the overall process of machine learning, and the common algorithms of machine learning.

Objectives

Upon completion of this course, you will be able to:

- Master the learning algorithm definition and machine learning process.
- Know common machine learning algorithms.
- Understand concepts such as hyperparameters, gradient descent, and cross validation.

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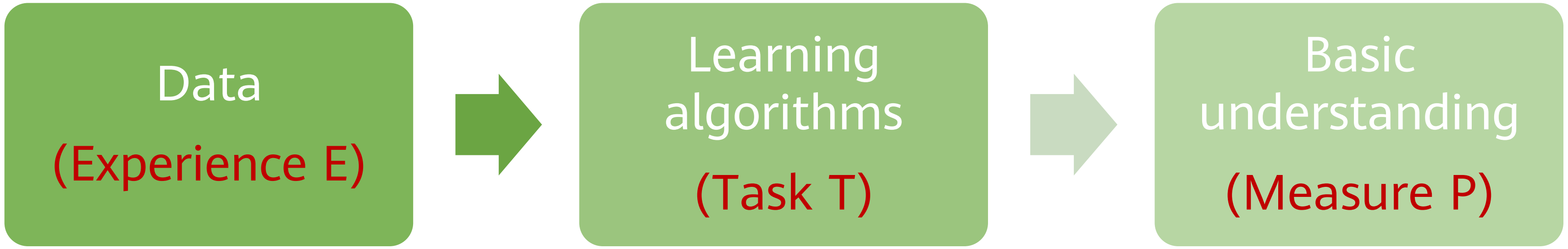
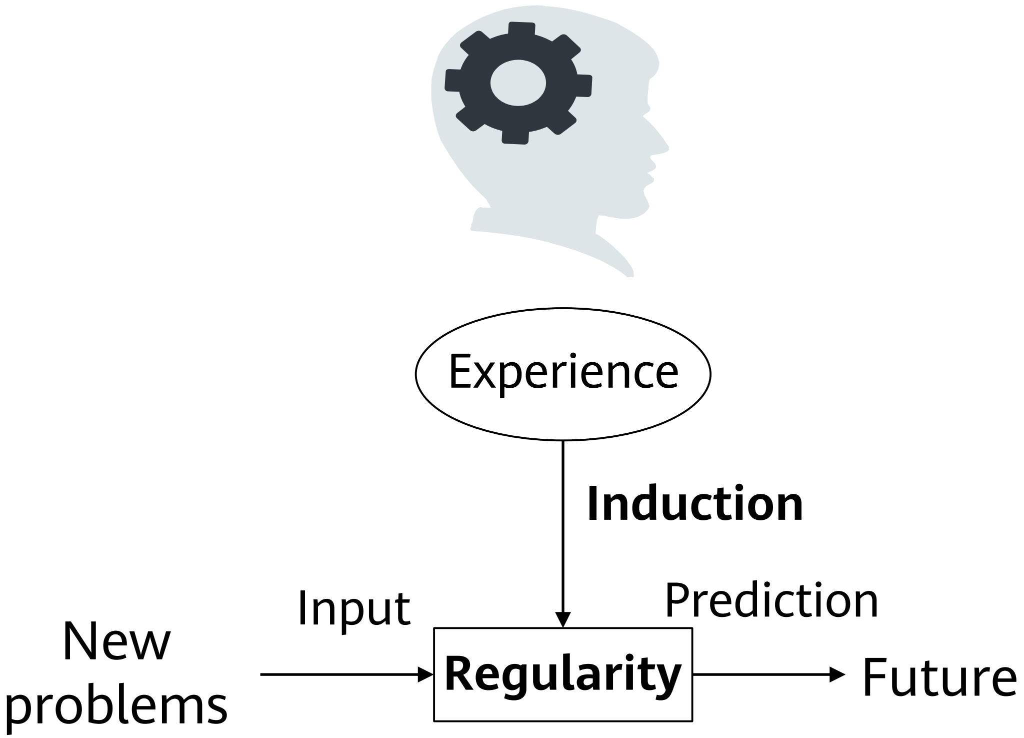
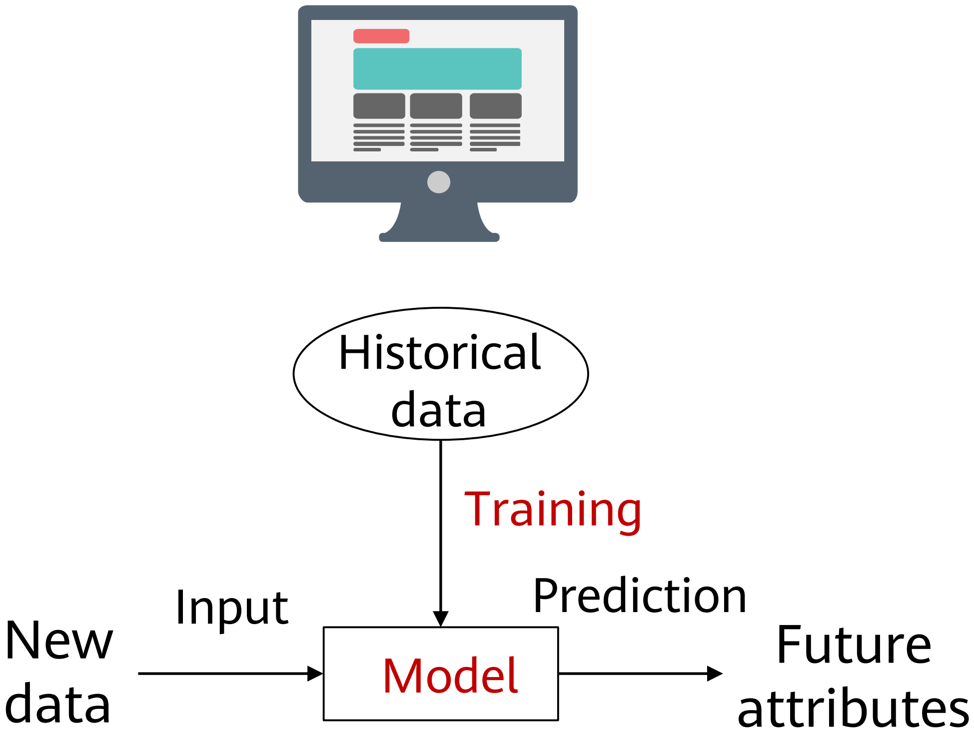
Machine Learning Algorithms (1)

- Machine learning (including deep learning) is a study of learning algorithms. A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T, as measured by P, improves with experience E.

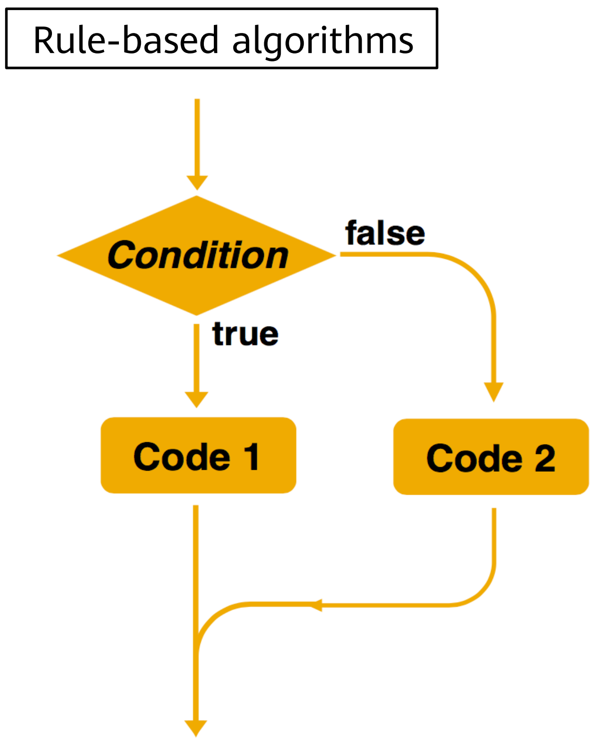


Machine Learning Algorithms (2)

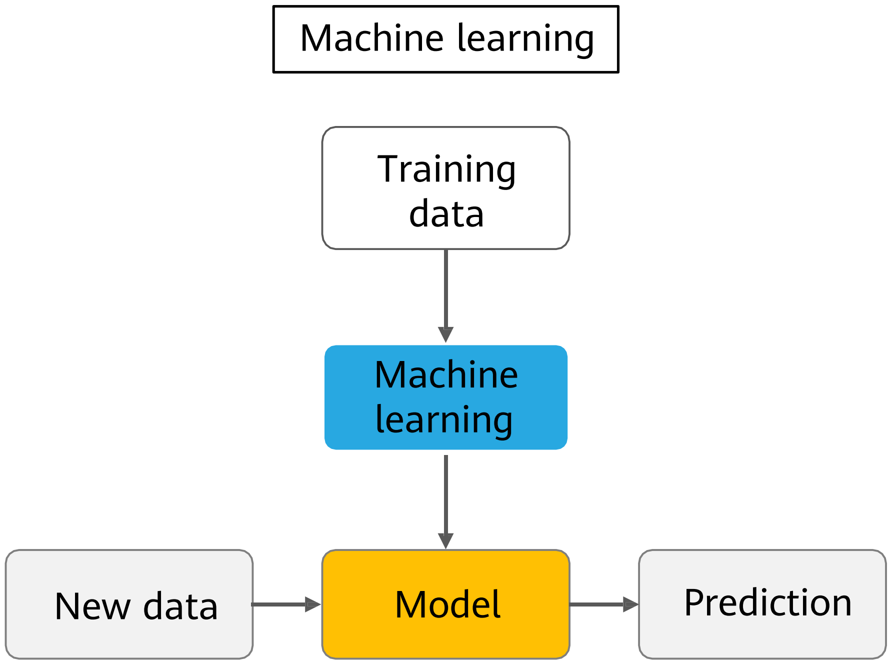




Differences Between Machine Learning Algorithms and Traditional Rule-Based Algorithms



- Explicit programming is used to solve problems.
- Rules can be manually specified.

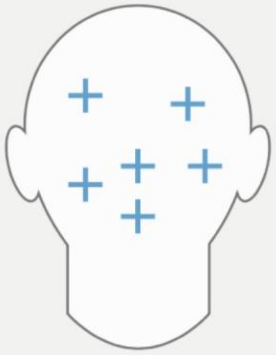


- Samples are used for training.
- The decision-making rules are complex or difficult to describe.
- Rules are automatically learned by machines.

Application Scenarios of Machine Learning (1)

- The solution to a problem is complex, or the problem may involve a large amount of data without a clear data distribution function.
- Machine learning can be used in the following scenarios:

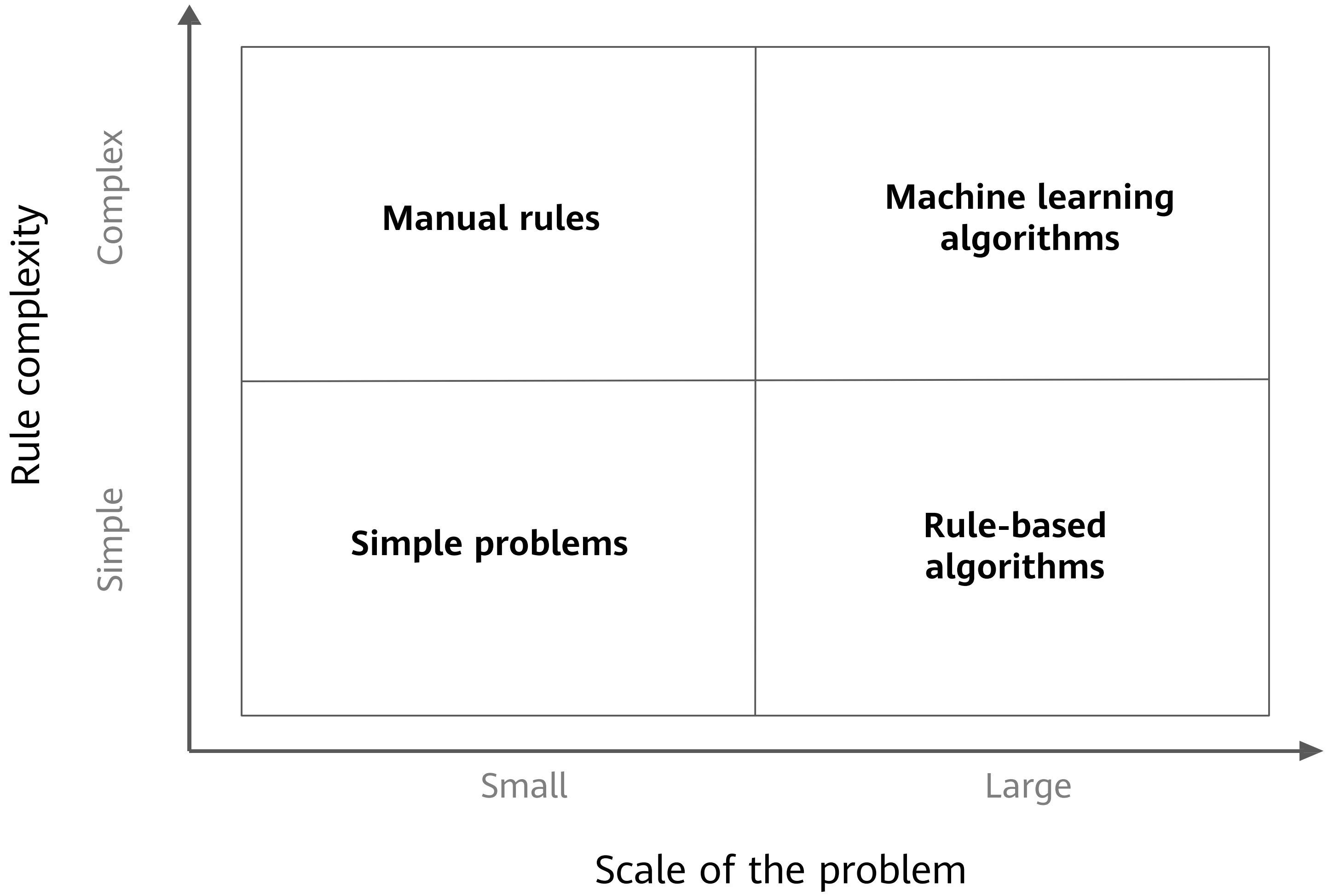
Rules are complex or cannot be described, such as facial recognition and voice recognition.



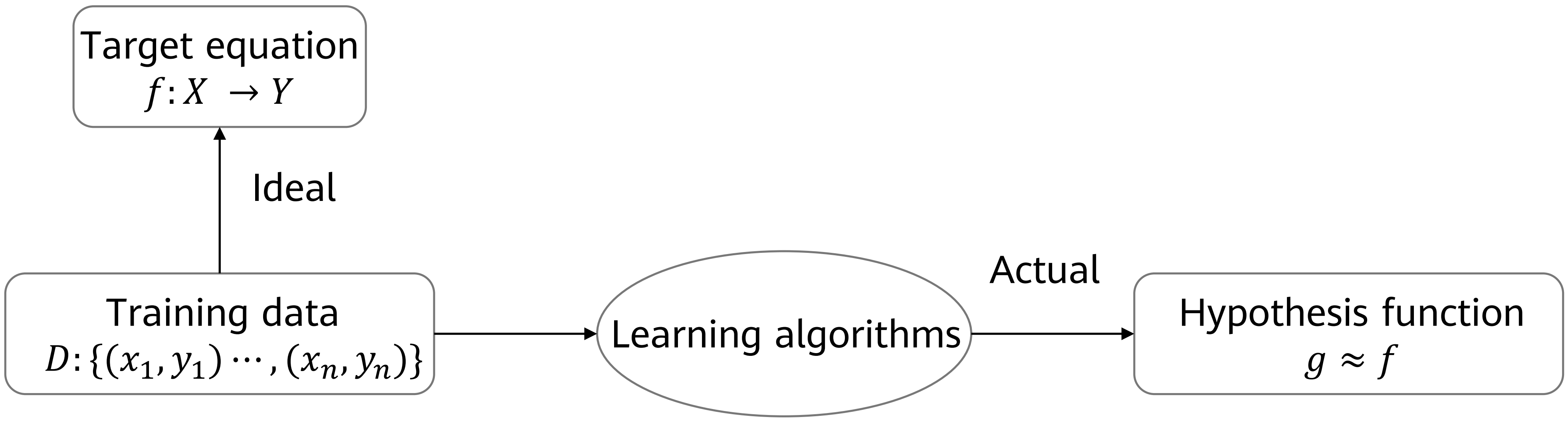
Task rules change over time. For example, in the part-of-speech tagging task, new words or meanings are generated at any time.

Data distribution changes over time, requiring constant readaptation of programs, such as predicting the trend of commodity sales.

Application Scenarios of Machine Learning (2)



Rational Understanding of Machine Learning Algorithms



- Target function f is unknown. Learning algorithms cannot obtain a perfect function f.
- Assume that hypothesis function g **approximates** function f, but may be different from function f.

Main Problems Solved by Machine Learning

- Machine learning can deal with many types of tasks. The following describes the most typical and common types of tasks.
  - Classification: A computer program needs to specify which of the k categories some input belongs to. To accomplish this task, learning algorithms usually output a function $f: R^n \rightarrow (1,2,\ldots,k)$. For example, the image classification algorithm in computer vision is developed to handle classification tasks.
  - Regression: For this type of task, a computer program predicts the output for the given input. Learning algorithms typically output a function $f: R^n \rightarrow R$. An example of this task type is to predict the claim amount of an insured person (to set the insurance premium) or predict the security price.
  - Clustering: A large amount of data from an unlabeled dataset is divided into multiple categories according to internal similarity of the data. Data in the same category is more similar than that in different categories. This feature can be used in scenarios such as image retrieval and user profile management.
- Classification and regression are two main types of prediction, accounting from 80% to 90%. The output of classification is discrete category values, and the output of regression is continuous numbers.

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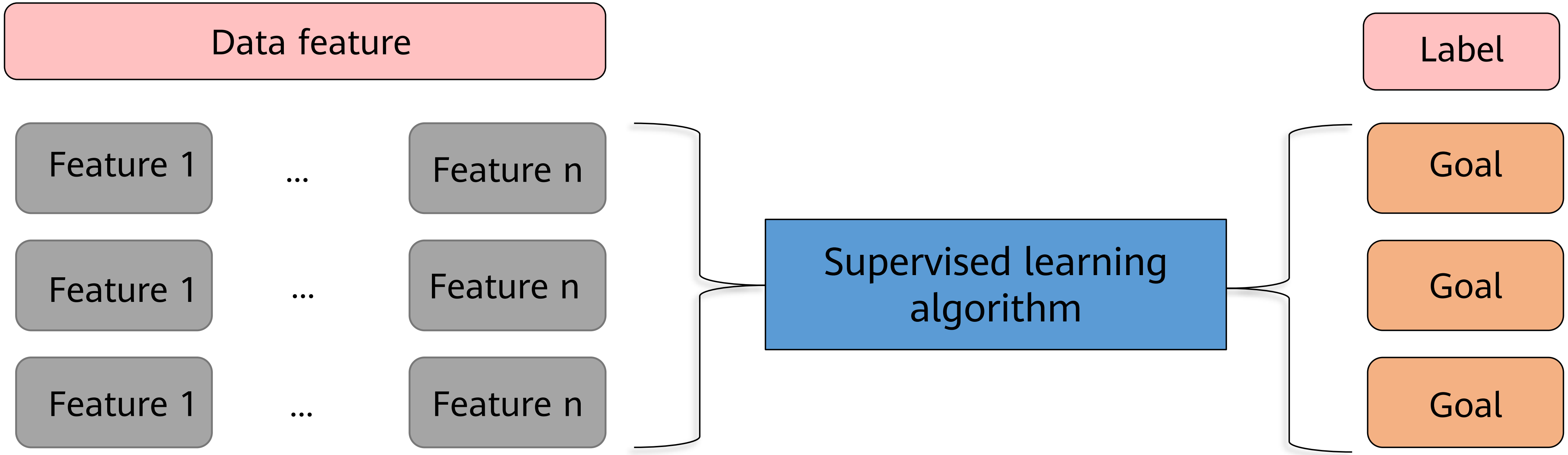
Machine Learning Classification

- **Supervised learning**: Obtain an optimal model with required performance through training and learning based on the samples of known categories. Then, use the model to map all inputs to outputs and check the output for the purpose of classifying unknown data.
- **Unsupervised learning**: For unlabeled samples, the learning algorithms directly model the input datasets. Clustering is a common form of unsupervised learning. We only need to put highly similar samples together, calculate the similarity between new samples and existing ones, and classify them by similarity.
- **Semi-supervised learning**: In one task, a machine learning model that automatically uses a large amount of unlabeled data to assist learning directly of a small amount of labeled data.
- **Reinforcement learning**: It is an area of machine learning concerned with how agents ought to take actions in an environment to maximize some notion of cumulative reward. The difference between reinforcement learning and supervised learning is the teacher signal. The reinforcement signal provided by the environment in reinforcement learning is used to evaluate the action (scalar signal) rather than telling the learning system how to perform correct actions.

Supervised Learning

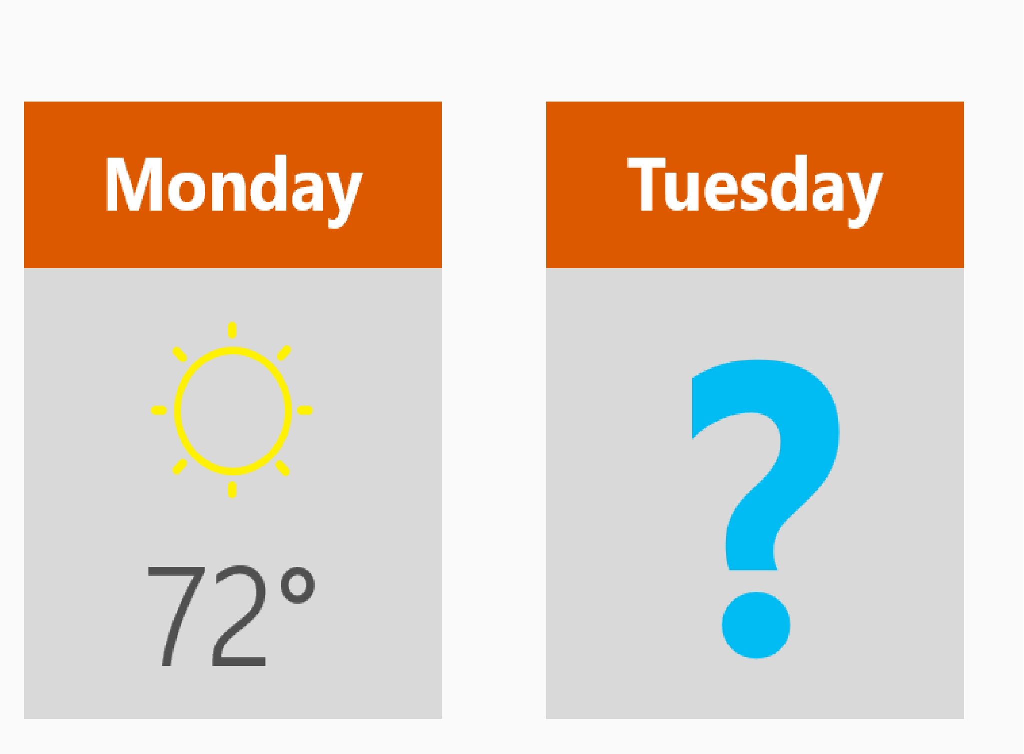


Weather	Temperature	Wind Speed
Sunny	Warm	Strong
Rainy	Cold	Fair
Sunny	Cold	Weak

Enjoy Sports
Yes
No
Yes

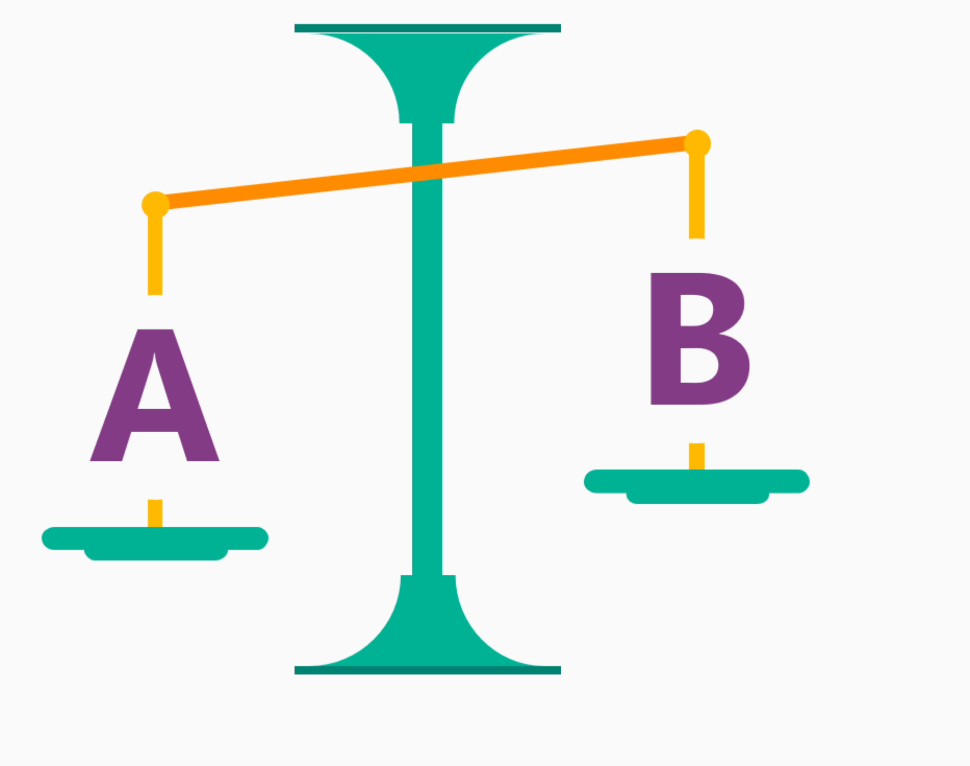
Supervised Learning - Regression Questions

- Regression: reflects the features of attribute values of samples in a sample dataset. The dependency between attribute values is discovered by expressing the relationship of sample mapping through functions.
  - How much will I benefit from the stock next week?
  - What's the temperature on Tuesday?

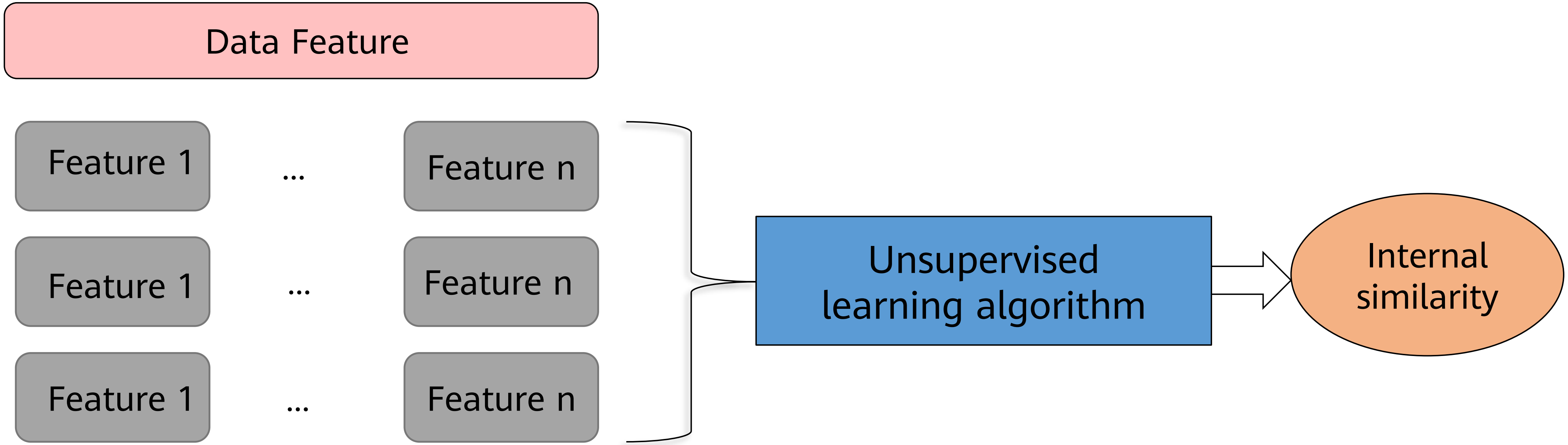


Supervised Learning - Classification Questions

- Classification: maps samples in a sample dataset to a specified category by using a classification model.
  - Will there be a traffic jam on XX road during the morning rush hour tomorrow?
  - Which method is more attractive to customers: 5 yuan voucher or 25% off?



Unsupervised Learning

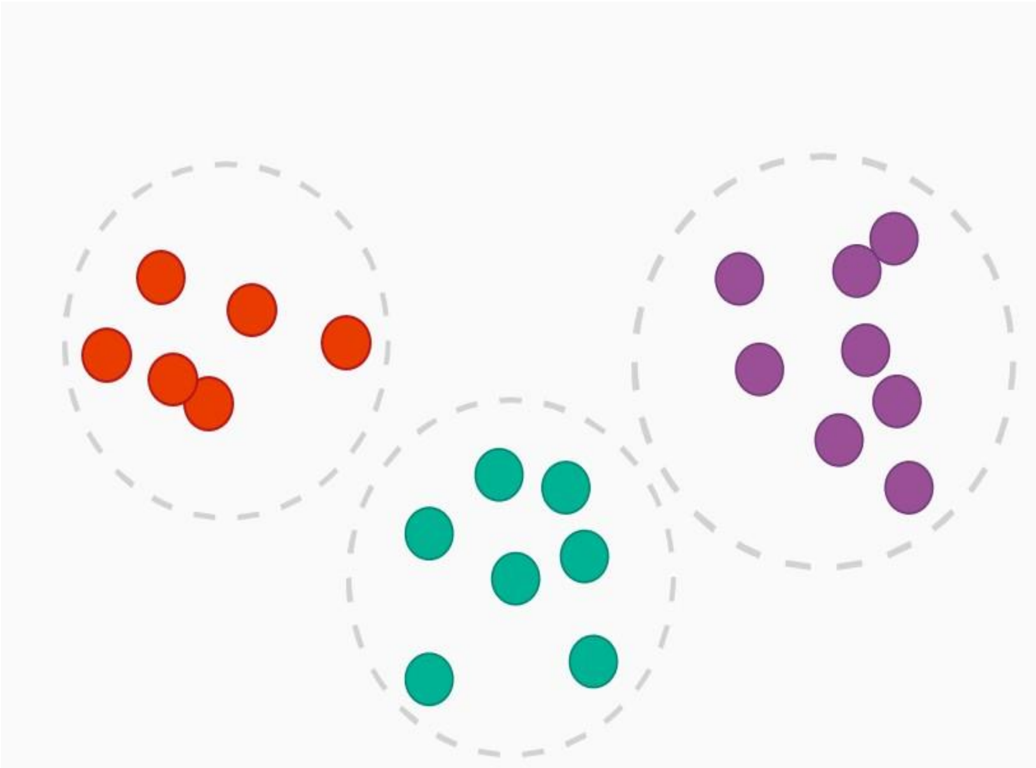


Monthly Consumption	Commodity	Consumption Time
1000–2000	Badminton racket	6:00–12:00
500–1000	Basketball	18:00–24:00
1000–2000	Game console	00:00–6:00

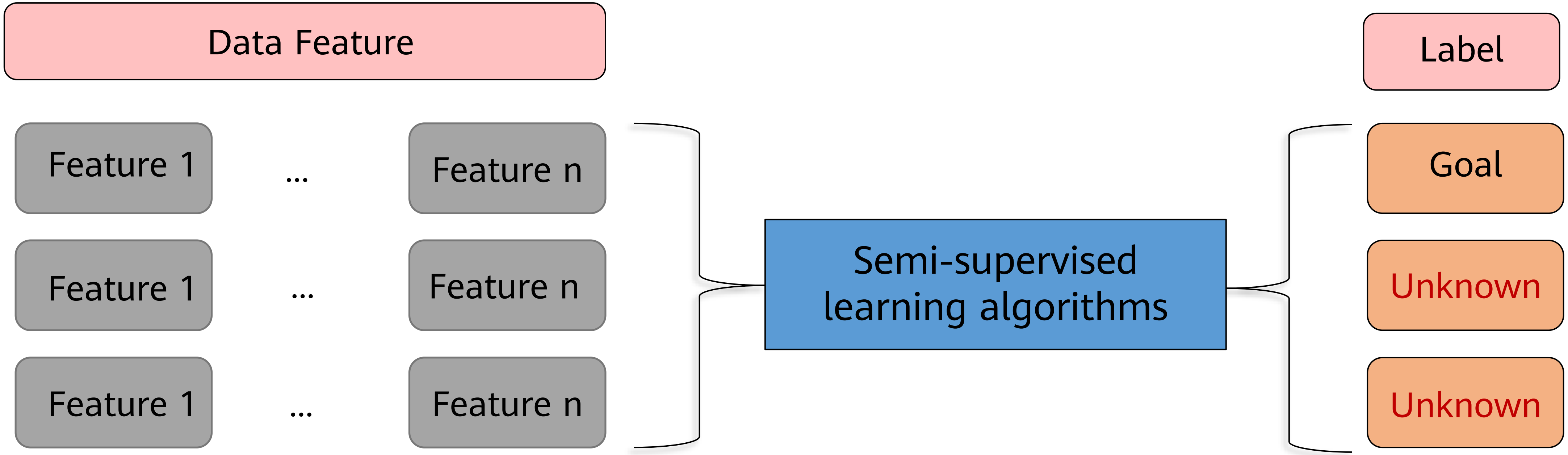
Category
Cluster 1
Cluster 2

Unsupervised Learning - Clustering Questions

- Clustering: classifies samples in a sample dataset into several categories based on the clustering model. The similarity of samples belonging to the same category is high.
  - Which audiences like to watch movies of the same subject?
  - Which of these components are damaged in a similar way?



Semi-Supervised Learning



Weather	**Temperature**	**Wind Speed**
Sunny	Warm	Strong
Rainy	Cold	Fair
Sunny	Cold	Weak

Enjoy Sports
Yes
/
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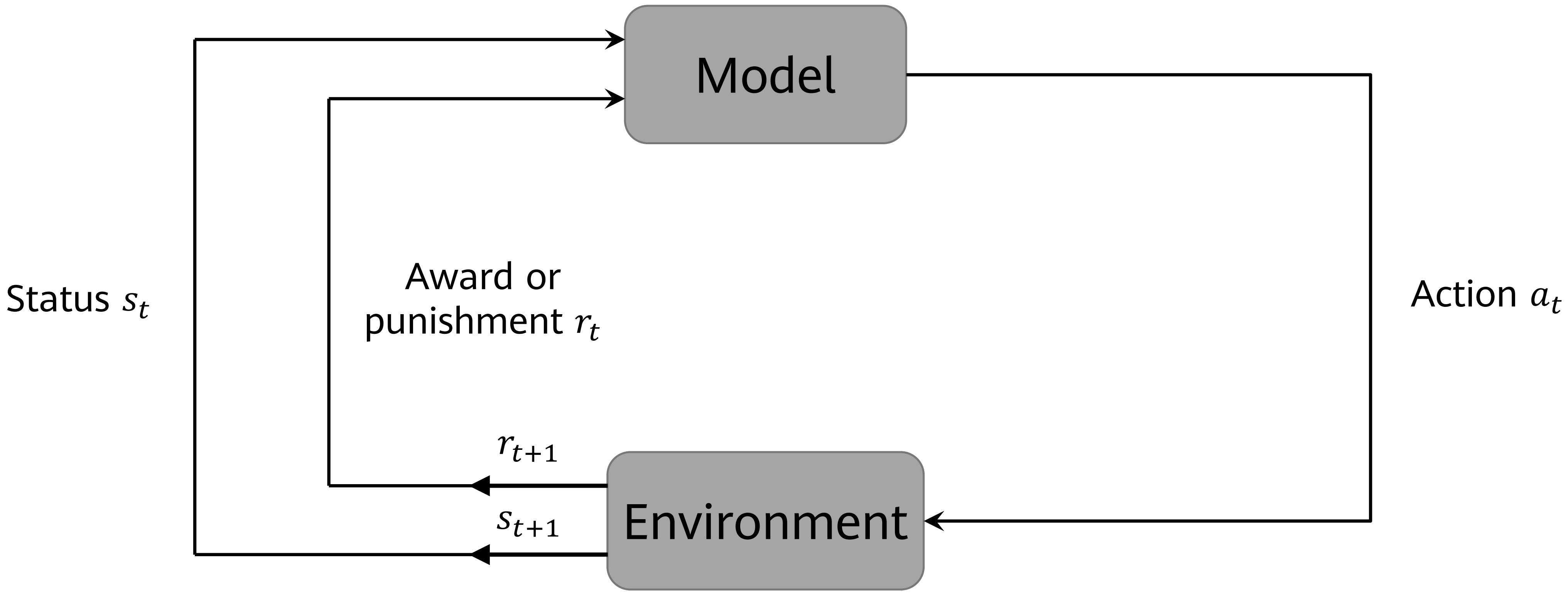
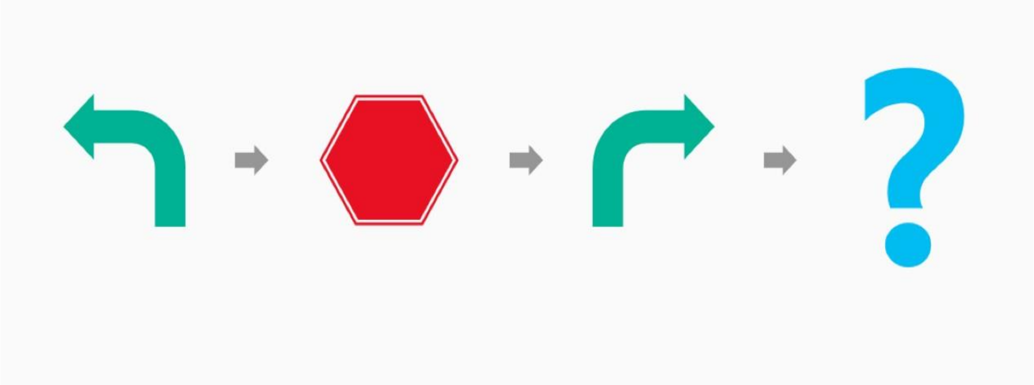
Reinforcement Learning

- The model perceives the environment, takes actions, and makes adjustments and choices based on the status and award or punishment.



Reinforcement Learning - Best Behavior

- Reinforcement learning: always looks for best behaviors. Reinforcement learning is targeted at machines or robots.
  - Autopilot: Should it brake or accelerate when the yellow light starts to flash?
  - Cleaning robot: Should it keep working or go back for charging?



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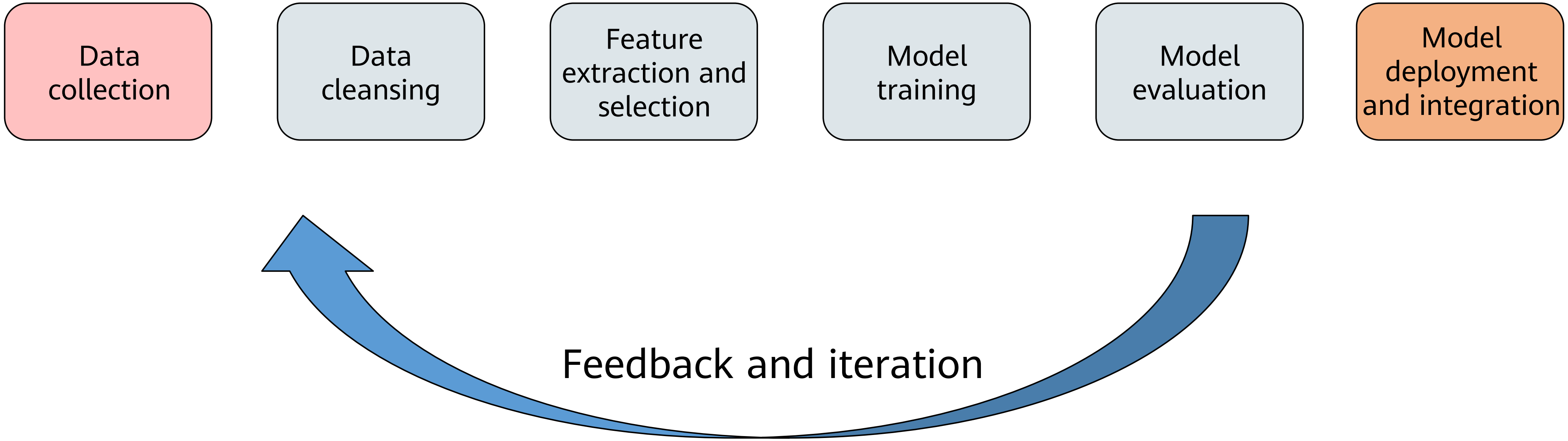
Machine Learning Process



Basic Machine Learning Concept — Dataset

- Dataset: a collection of data used in machine learning tasks. Each data record is called a sample. Events or attributes that reflect the performance or nature of a sample in a particular aspect are called features.
- Training set: a dataset used in the training process, where each sample is referred to as a training sample. The process of creating a model from data is called learning (training).
- Test set: Testing refers to the process of using the model obtained after learning for prediction. The dataset used is called a test set, and each sample is called a test sample.

Checking Data Overview

- Typical dataset form

		Feature 1	Feature 2	Feature 3	Label
	No.	Area	School Districts	Direction	House Price
Training set	1	100	8	South	1000
	2	120	9	Southwest	1300
	3	60	6	North	700
	4	80	9	Southeast	1100
Test set	5	95	3	South	850

Importance of Data Processing

- Data is crucial to models. It is the ceiling of model capabilities. Without good data, there is no good model.

Data preprocessing

Data cleansing

Fill in missing values, and detect and eliminate causes of dataset exceptions.

Data dimension reduction

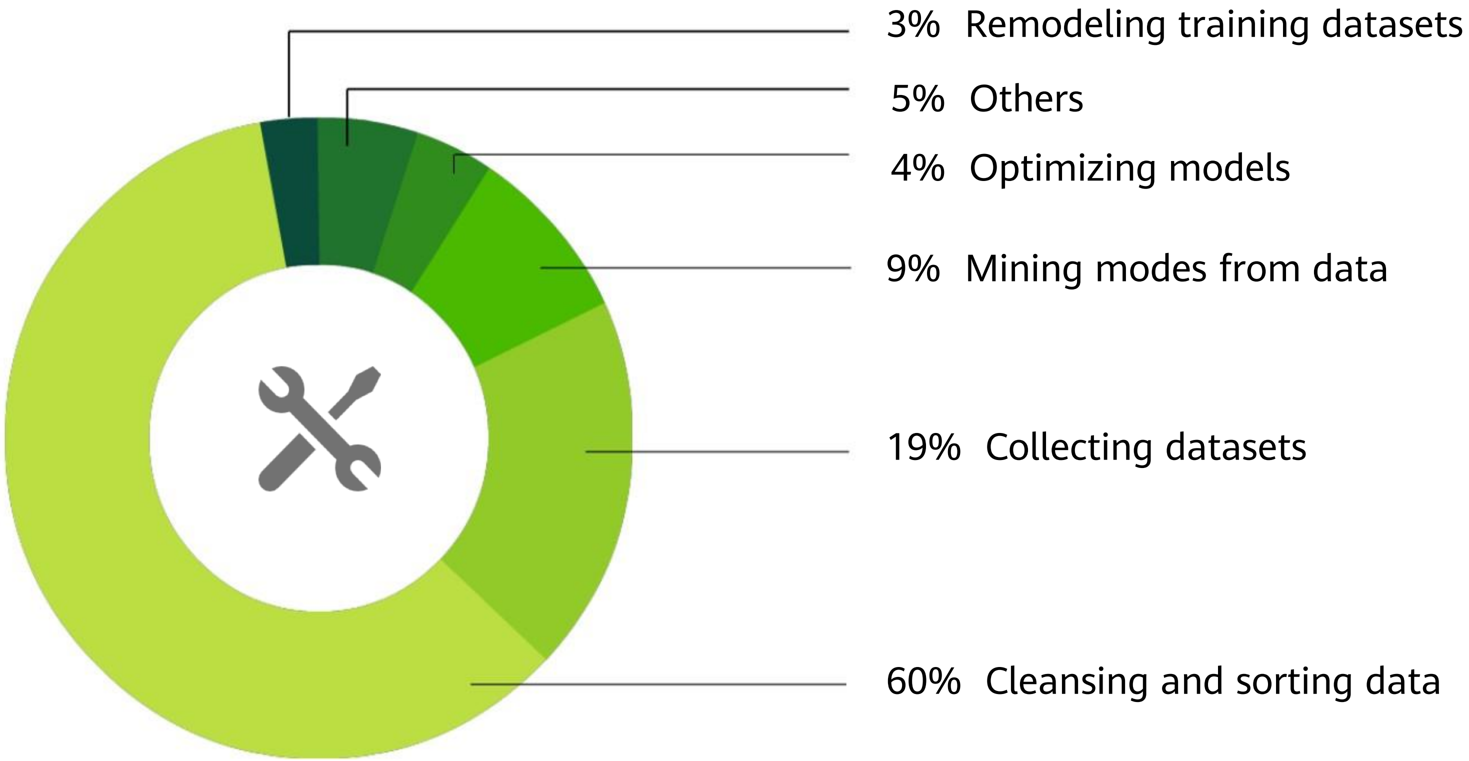
Simplify data attributes to avoid dimension explosion.

Data normalization

Normalize data to reduce noise and improve model accuracy.

Workload of Data Cleansing

- Statistics on data scientists' work in machine learning



CrowdFlower Data Science Report 2016